Epic 3: Data Pre-processing

Story 3: Handling Categorical Values

Machine learning algorithms require all input features to be in numerical format. Therefore, any categorical variables present in the flood prediction dataset must be converted into numerical values before model training. This transformation ensures that the algorithms can correctly interpret and process the data.

In this project, categorical features are converted into numerical representations using **Feature Mapping** and **Label Encoding** techniques.

Feature Mapping

Feature Mapping is applied when categorical values have a natural order or ranking. Each category is manually assigned a numerical value based on its significance or sequence.

Example:

- Low = 0
- Medium = 1
- High = 2

This method preserves the ordinal relationship between categories and makes the data suitable for machine learning models.

Label Encoding

Label Encoding is used for categorical variables containing multiple unique categories without any inherent order. The **LabelEncoder** class from the **Scikit-learn** library assigns a unique integer value to each category.

Example:

- Male = 0
- Female = 1

or

- Working = 0
- Student = 1
- Pensioner = 2

After label encoding, all categorical columns are transformed into machine-readable numerical values, enabling efficient model training and prediction.

Importance of Handling Categorical Values

- Converts text-based categories into numerical values.
- Improves compatibility with machine learning algorithms.
- Ensures accurate and efficient model training.
- Prevents errors caused by non-numeric data.
- Produces a fully prepared dataset for model development and evaluation.

Outcome

- Identified categorical features in the dataset.
- Applied Feature Mapping and Label Encoding techniques.
- Converted all categorical variables into numerical format.
- Prepared the dataset for machine learning model training and testing.

Figure 9: Converting categorical features into numerical values using Feature Mapping and Label Encoding.

```
#independent features
x=dataset.iloc[:,2:7].values
```

```
#dependent feature
y=dataset.iloc[:,9:].values
```